

Homework (Jalapeno)

Q1. The graph of $y = |x - 3|$ has a vertex at

- (A) (3, 0)
- (B) (0, 3)
- (C) (3, 3)
- (D) (0, 0)
- (E) (1, 1)

Q2. The ancient Egyptian architect Imhotep designed a pyramid with a base length of $|x + 5|$ meters. If the length is 10 meters, what is x ?

- (A) -15
- (B) -10
- (C) -5
- (D) 0
- (E) 5

Q3. The graph of $y = |2x - 1|$ has an axis of symmetry at

- (A) $x = 0$
- (B) $x =$

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- (B) $x = 1$
- (C) $x = 2$
- (D) $x = 3$

Q4. The ancient Greek philosopher Aristotle measured the distance between two points as $|x - 2| + |x + 1|$. If the distance is 5, what is x ?

- (A) -3
- (B) -2
- (C) -1
- (D) 0

(E) 1

Q5. The graph of $y = |x| + 2$ has a minimum value at

- (A) (0, 0)
- (B) (0, 2)
- (C) (1, 1)
- (D) (2, 3)
- (E) (3, 4)

Q6. The ancient Chinese mathematician Liu Hui graphed the function $y = |x - 2| - 1$. What is the direction of opening?

- (A) Up
- (B) Down
- (C) Left
- (D) Right
- (E) None

Q7. The ancient Babylonian astronomer Kidinnu measured the angle of the sun as $|x - 30|$ degrees. If the angle is 10 degrees, what is x ?

- (A) 20
- (B) 25
- (C) 30
- (D) 35
- (E) 40

Q8. The graph of $y = -|x + 1|$ has a maximum value at

- (A) (-1, 0)
- (B) (-1, 1)
- (C) (0, 0)
- (D) (0, 1)
- (E) (1, 1)

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 2| + 1$ and identify the vertex, axis of symmetry, and direction of opening. Show your work and explain your reasoning.

Q10. The ancient Mayan calendar measured time as $|x - 5| + |x + 2|$. If the time is 10, what is x ? Show your work and explain your reasoning.

Chef's Tips

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- Tip 1: Encourage students to use graph paper to visualize the functions.
- Tip 2: Remind students to check their work and explain their reasoning.
- Tip 3: Suggest that students use different colors to highlight important features of the graphs.